

IdeaStamp Assignment 3 Technical Note

Smart Contract Implementation, Unit Testing Plan, and dApp Wireframe Summary

1. Prototype Scope

This Assignment 3 package translates the submitted IdeaStamp white paper into a logically consistent, course-scope prototype.

The implementation focuses on the core workflow promised in the white paper: commit, reveal, verify, and an optional lightweight dispute registration module.

The prototype intentionally defers ERC20 token launch, DAO governance, staking-based arbitration, organization memberships, and advanced storage integrations to future work.

2. Contracts Included

Contract	Purpose	Key Functions
IdeaRegistry.sol	Main priority registry	commit, reveal, computeCommitment, getCommitment
DisputeRegistryLite.sol	Minimal dispute recording	openDispute, resolveDispute, getDispute

3. White Paper Alignment

White paper promise	Assignment 3 status
Timestamped commitment registry	Implemented
Commit-reveal verification	Implemented
Public verification workflow	Implemented
Dispute handling	Partially implemented as dispute-lite
IDEA token / staking / DAO	Deferred to future work

4. Testing Plan

IdeaRegistry tests cover successful commit, duplicate commit rejection, insufficient fee rejection, successful reveal, reveal of unknown commitment, double reveal rejection, author commitment tracking, and owner-only fee updates.

DisputeRegistryLite tests cover dispute creation with valid commitments, rejection of invalid disputes, and admin-side resolution.

A testing evidence template is included in docs/testing_evidence_template.md so real logs can be pasted after running the suite in Cursor or on a local machine.

5. Tooling and Commands

Framework: Hardhat

Solidity compiler target: 0.8.24

Suggested commands: npm install, npm run compile, npm test, npm run deploy:local

6. Deferred Features

- ERC20 IDEA token and incentive distribution
- staking/slashing for dispute arbitration
- DAO governance and treasury spending
- organization workspaces and memberships
- encrypted storage gateway and advanced integrations

7. Submission Notes

This package already contains the core source code, unit tests, wireframe PDF, README, mapping note, and a Cursor execution guide.

The only missing evidence that must be produced outside this container is the real compile/test terminal log, because this environment cannot install the required Hardhat npm packages.